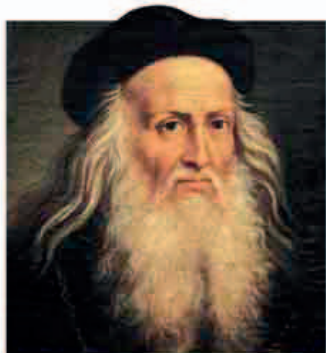


“ALTHOUGH NATURE COMMENCES WITH REASON AND ENDS IN EXPERIENCE IT IS NECESSARY FOR US TO DO THE OPPOSITE...”

Leonardo da Vinci, Italian painter, architect, and engineer, in *Notebooks*

THE LATE 15TH CENTURY

saw the production of the first printed practical mathematical textbooks. The *Treviso Arithmetic*, printed in 1478, demonstrated techniques of **addition** and



LEONARDO DA VINCI
(1452–1519)

Born in Tuscany, Leonardo da Vinci was the most creative mind of the Renaissance. He became an apprentice sculptor before moving to Milan to work for the ruling Sforza family. A talented artist, his paintings include *The Last Supper* (1495–98) and *Mona Lisa* (c. 1503). His scientific interests were vast and he produced 13,000 pages of notebooks.

Leonardo da Vinci's notebooks

This page from Leonardo's notebooks shows his sketches of flying machines. He used mirror-writing to make notes, although it is uncertain why he did so.

subtraction, as well as **division** and five different ways of performing **multiplication**, including cross-multiplication and a “chessboard” technique similar to modern practice. It also dealt with the rule of mixtures (for instance, showing the proportions of a precious metal in alloys) and **methods of calculating the Golden Number** (see 1723–24).

In 1483, it was followed by a German counterpart, the *Bamberger Rechenbuch* (*Bamberger Arithmetic*), which, as well as setting out five procedures for multiplication, gave **rules for the summing of geometrical and arithmetical progressions**.

The prolific scientific interests of the Italian painter, architect, and engineer Leonardo da Vinci led him to his studies on the mechanisms of flight. He considered that “the bird is an instrument operating through mathematical laws” and **he worked on designs for flying machines** using birdlike wings.

In 1481, he also devised a **parachute**, with a sealed linen cloth supported by wooden poles that made a pyramidal shape, that would reduce the rate of acceleration in a fall and cushion the wearer's impact. There is, however, no evidence that Leonardo actually built any of these fantastic machines.

THE DEVELOPMENT OF MATHEMATICAL NOTATION

progressed rapidly during the late 15th century. The *Dresden Manuscript* (1461) gave **special symbols for the first four powers of x**, and, around 1489, the German mathematician Johannes Widman (1462–98) wrote the first work to use the signs “+” and “–” to represent the operations of addition and subtraction. He also used a long line to represent “equals.”

Around 1489, Leonardo **began a study of human anatomy**, using dissections of animals and human corpses (he claimed to have dissected 10). He recorded his findings in notebooks between 1489 and 1507. In these,

“I... STEERED FOR THE CANARY ISLANDS... THENCE TO TAKE MY DEPARTURE AND PROCEED TILL I ARRIVED AT THE INDIES.”

Christopher Columbus, from *Journal of the First Voyage*, 1492

he made the most detailed anatomical drawings yet seen.

In 1490, Leonardo was **the first person to describe capillary action**, the ability of water in tiny spaces to “crawl upward,” acting contrary to other natural forces (such as gravity).

In October 1492, the Genoese explorer Christopher Columbus

landed at San Salvador in the Bahamas, **the first European to reach the Americas** since the Vikings in the 11th century. His voyage led to an exchange of population, food crops, and diseases as well as the discovery of large numbers of hitherto unseen species, such as the llama and armadillo.

NAVIGATING AND MAPPING THE WORLD



The translation of Ptolemy's *Geography* from Greek to Latin in 1409 and the Portuguese voyages down the west coast of Africa gave an impetus to map-making techniques.

Maps of the 15th century, such as this 1540 map (left) by Venetian monk and mapmaker Fra Mauro, combined a knowledge derived from Ptolemy with information sourced from mariners' charts, but did not use a projection that portrayed distances accurately. It wasn't until 1569 and Flemish geographer and cartographer Gerardus Mercator's world map that maps really helped sailors determine routes at sea more easily.

1481 Leonardo da Vinci produces a design for a parachute

1489 Leonardo da Vinci begins series of anatomical drawings based on dissections

1492 Christopher Columbus lands in the Bahamas

1489 A treatise by Johannes Widman is first to use “+” and “–” signs

1490 Leonardo describes the principle of capillary action

1492 Italian scholar Ermolao Barbaro suggests *Plinianae* thousands of corrections to Pliny's *Natural History*

1494 Italian mathematician Luca Pacioli's *Algebra*—the first comprehensive work on the subject—is printed

1496 Regiomontanus's *Epitome of Ptolemy's Almagest* is published, making his astronomical theories more accessible